**Exam 3 –Practice 1.**

*You may use a basic calculator and formula sheets. Your sheets must contain formulas only, no examples. Phones and all other smart devices are forbidden. Any AI usage will result in an automatic F for the exam. Show **ALL** your work for credits.*

1. Solve each exponential equation.
   a. $9^x = 7$

$$9^x = 7$$

$$\log 9^x = \log 7$$

$$x \log 9 = \log 7$$

$$x = \frac{\log 7}{\log 9} \approx .88562$$

   b. $6 \cdot 7^{4x+1} - 11 = 5$

$$\Leftrightarrow \quad 6 \cdot 7^{4x+1} = 5 + 11$$

$$\Leftrightarrow \quad 6 \cdot 7^{4x+1} = 16$$

$$\Leftrightarrow \quad 7^{4x+1} = \frac{16}{6}$$

$$\Leftrightarrow \quad \log 7^{4x+1} = \log\left(\frac{16}{6}\right)$$

$$\Leftrightarrow \quad (4x+1)\cdot \log 7 = \log\left(\frac{16}{6}\right)$$

$$\Leftrightarrow \quad 4x+1 = \frac{\log\left(\frac{16}{6}\right)}{\log 7} \approx .5040$$

$$\Leftrightarrow \quad 4x = .5040 - 1$$

$$\Leftrightarrow \quad x = \frac{.5040 - 1}{4} = -.124$$

c. $7^{4x+1} = 9^{2x+3}$

$$\log\left(7^{4x+1}\right) = \log\left(9^{2x+3}\right)$$

$$\Leftrightarrow \quad (4x+1)\cdot \log 7 = (2x+3)\cdot \log 9$$

$$\Leftrightarrow \quad (4x+1)\cdot .845 = (2x+3)\cdot .954$$

$$\Leftrightarrow \quad 4 \cdot .845\, x + .845 = 2 \cdot .954\, x + 3 \cdot .954$$

$$\Leftrightarrow \quad 3.38x + .845 = 1.908x + 2.862$$

$$\Leftrightarrow \quad 3.38x - 1.908x = 2.862 - .845$$

$$\Leftrightarrow \quad 1.472x = 2.017$$

$$\Leftrightarrow \quad x = \frac{2.017}{1.472} \approx \boxed{1.37}$$

2. The number of bachelor's business degrees (in thousands) conferred in the United States can be modeled by an exponential function. In 2010, 279,000 business degrees were conferred, and the number of degrees has risen 2% each year.

   a. Write an equation to model the number of business degrees conferred.

$$y = a.(1+r)^t \quad \text{(Exponential Growth formular)}$$

$r = 2\%$

$t$ : the number of years since 2010 (in 2010, $t = 0$)

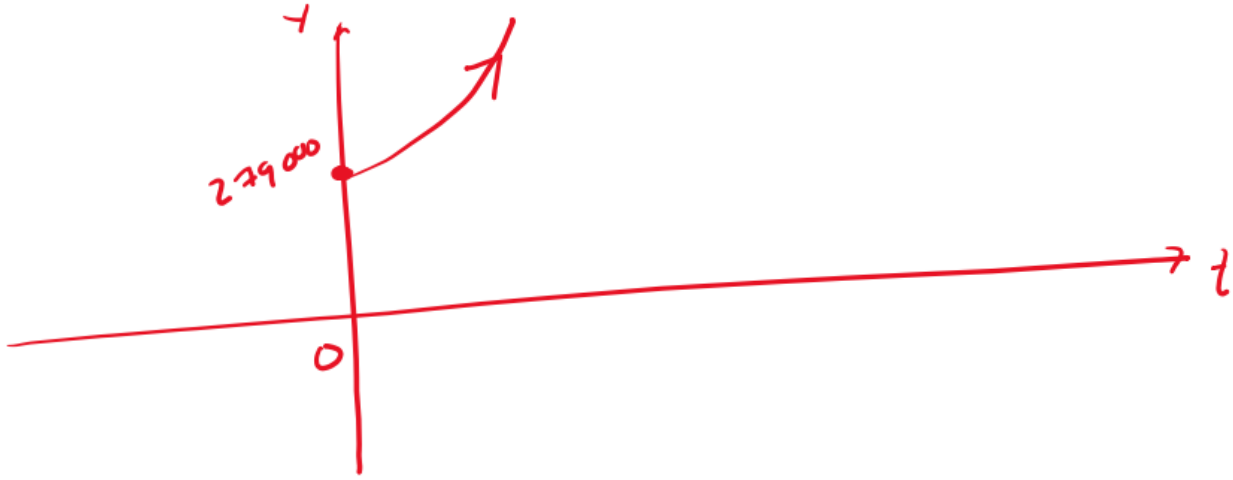
$a = 279,000$

$y$ : the number of business degrees conferred

$$\Rightarrow \quad y = 279,000(1 + .02)^t$$

$$y = 279,000 * (1.02)^t$$

b. Graph the equation




c. What was the first year in which 475,000 business degrees were conferred?

we need to find t such that

$$y = 475,000$$

$$279,000 \times (1.02)^t = 475,000$$

$$\Leftrightarrow \quad (1.02)^t = \frac{475,000}{279,00}$$

$$\Leftrightarrow \quad (1.02)^t \approx 1.703$$

$$\Leftrightarrow \quad \log 1.02^t = \log 1.703$$

$$\Leftrightarrow \quad t \cdot \log 1.02 = \log 1.703$$

$$\Leftrightarrow \quad t = \frac{\log 1.703}{\log 1.02} \approx 26.88$$

Round down : $t = 26$.

The year we need to find is $2010 + 26 = 2036$

d. Use the model to estimate how many degrees will be conferred in 2027.

In 2027, $t = 2027 - 2010 = 17$

$$y = 279{,}000 \cdot (1.02)^{17} \approx 390{,}667$$

**3.** A piece of machinery is purchased for $40,000. The business estimates that its value will decrease 43% each year.

a. Write an equation to model the value of the machinery after *t* years.

Exponential Decay formula:

$$y = a(1-r)^t$$

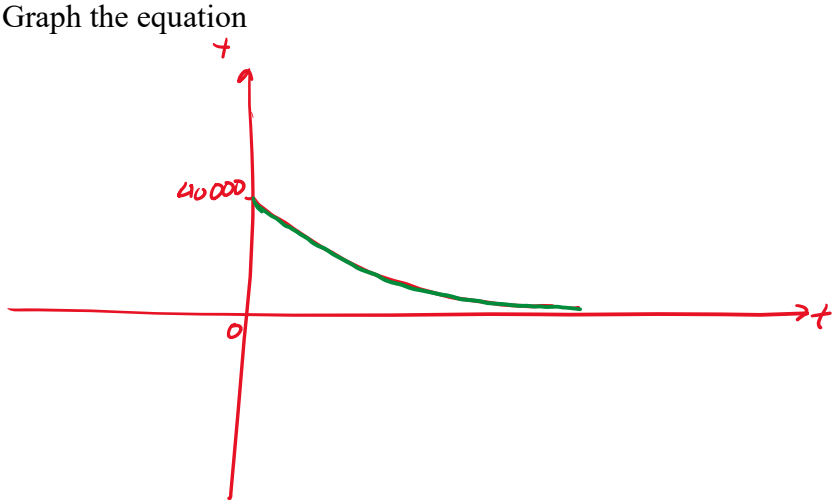
$a = 40{,}000$

$r = 43\% = .43$

$$\Rightarrow \quad y = 40{,}000\,(1-.43)^t$$

$$y = 40{,}000 \cdot (.57)^t$$

b. Graph the equation



c. What is the value of the machinery after 2 years?

$t = 2$

$\Rightarrow \quad y = 40{,}000 \cdot (.57)^2 = 12{,}996$

d. When will the machinery be worth $5,000?

$y = 5000$

$\Rightarrow \quad 40{,}000 \cdot (.57)^t = 5000$

$\Rightarrow \quad (.57)^t = \dfrac{5000}{40000} = .125$

$\Rightarrow \quad \log .57^t = \log .125$

$\Rightarrow \quad t \cdot \log .57 = \log .125$

$\Rightarrow \quad t = \dfrac{\log(.125)}{\log(.57)} = 3.6993$

4. A rookie player in the NFL just signed his first 7-year contract. To prepare for his future, he deposits \$150,000 at the end of each year for 7 years in an account with 3.1% compounded annually.

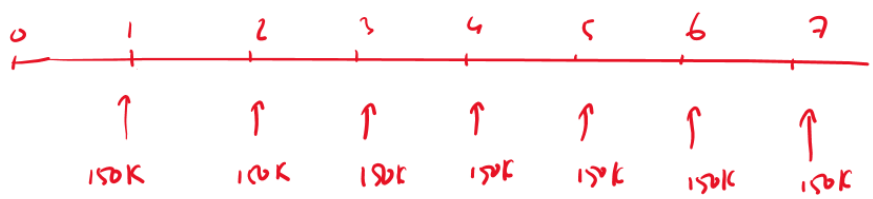
a. How much will he have in his account after 7 years?

**The Formula**

$$FV = P \times \left[\frac{(1+r)^n - 1}{r}\right]$$

**Where:**

- $FV$: The Future Value (the total amount you want to have).
- $P$: The Periodic payment (the amount you deposit each period).
- $r$: The interest rate per period (expressed as a decimal).
- $n$: The total number of periods (e.g., total years or months).




$P = 150,000$

$r = 3.1\%$

$n = 7$

$$FV = P \cdot \left[\frac{(1+r)^n - 1}{r}\right] = 150,000 \cdot \left[\frac{1.031^7 - 1}{.031}\right]$$

$$= 1,152,834.59$$

b. How much interest was earned over the 7 years?

Total deposits : 150,000 × 7

Interest earned : 1152834.59 − 150,000 × 7

= 102,834.59

5. A wealthy family member has funded an annuity that will pay you $1500 at the end of each year for six years. If the annual interest rate is 8%, compounded annually, find the present value of the annuity.

**The Formula**

$$PV = P \times \left[\frac{1-(1+r)^{-n}}{r}\right]$$

**Where:**

- $PV$: The Present Value (the lump sum you start with today).
- $P$: The Periodic payment (the amount you receive each period).
- $r$: The interest rate per period (as a decimal).
- $n$: The total number of periods.

$$PV = P \cdot \left[ \frac{1 - (1+r)^{-n}}{r} \right]$$

$P = 1500$

$r = .08$

$n = 6$

$$\Rightarrow \quad PV = 1500 \cdot \left[ \frac{1 - (1+.08)^{-6}}{.08} \right]$$

$$= 6934.319$$

6. The average used car price is $32,000. The average annual interest rate is currently 6.39% for a 60-month loan (5 years). If you have $5,000 for a down payment,

a. what can you expect the monthly payment to be?

Formula:

Amortization is the process of paying off a debt over a set period through a series of regular, fixed payments.

$$M = L \times \left[ \frac{r(1+r)^n}{(1+r)^n - 1} \right]$$

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**Variable Guide**

- $M$: Monthly Payment (The amount you pay each period).
- $L$: Loan Amount (The total principal you are borrowing today).
- $r$: Monthly Interest Rate (Annual Rate / 12).
- $n$: Total Number of Payments (Loan Term in years $\times$ 12).

$$M = L \times \left[ \frac{r(1+r)^n}{(1+r)^n - 1} \right]$$

$L = 32{,}000 - 5000 = 27{,}000$

$n = 60$

$r = 6.39\% / 12 = .005325$

$$M = 27000 * \left[ \frac{.005325 \cdot (1 + .005325)^{60}}{(1 + .005325)^{60} - 1} \right]$$

$= 526.896$

b. How much will you pay in total for the car?

Total = down payment + 526.896 × 60

= 5000 + 526.896 × 60 = $\boxed{36{,}613.76}$

7. In 2023, the average amount of student loan debt for a bachelor's degree recipient at a private university was $54,921. Assume an annual interest rate of 6.87%

a. Find the monthly payment and calculate the total interest paid over the course of the loan for a 10-year payment plan.

Formula :

$$M = L \times \left[ \frac{r(1+r)^n}{(1+r)^n - 1} \right]$$

L = Loan = 54,921

$r = \frac{\text{annual rate}}{12} = \frac{6.87\%}{12} = .005725$

n = loan term × 12 = 10 × 12 = 120

⇒ M = 634.0057

Monthly payment, M = 634.0057

Interest = Total monthly payments − Loan

Interest = 634.0057 × 120 − 54,921 = 21,166.2

b. Find the monthly payment and calculate the total interest paid over the course of the loan for a 20-year payment plan.

$n = 20 \times 12 = 240$

$r = .005725$

$\Rightarrow \quad M = 421.5269$

$\text{Interest} = 421.5269 \times 240 - 54{,}921$

$= 46{,}245.45$